Project 2

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1. Zipf Distribution Computation

- **Function:** After a search, the **"Analyze Search Results"** button appears.
- **Action:** Clicking it gathers all text from the displayed documents, tokenizes it (converts to lowercase words), and calculates the frequency of each word.
- **Result:** The Zipf distribution is displayed in a pop-up modal containing a ranked table and a scatter plot of word frequencies.

2. Porter's Algorithm Implementation

- **Function:** The website includes a self-contained **Porter Stemmer** JavaScript module.
- **Action:** When the "Analyze" button is clicked, this module runs in parallel with the Zipf analysis. It processes the *exact same text*
- **Result:** A second, stemmed frequency distribution is generated.

3. Comparative Analysis Interface

The primary goal of the "Zipf Distribution Comparison" modal is to compare the two methods:

- **Side-by-Side Table:** Directly compares the ranked lists of original terms and stemmed terms, showing how word rankings and frequencies change.
- **Overlaid Chart:** The scatter plot displays both distributions on the same graph. Checkboxes allow the user to toggle each view, visually highlighting the difference in the curves.
- **Key Statistics:** At a glance, the user can see the difference in **Total Words** and **Unique Terms** between the two methods, demonstrating the immediate impact of stemming.